

Trends in Smartphone Screen Time and Its Impact on Sleep: Analysis Using Social Media Usage and Sleep Data in Singapore

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Abstract—Sleep habits have changed globally as a result of smartphones’ widespread integration into daily life; this phenomenon is especially noticeable in densely populated cities like Singapore. Unquestionably, “always-on” connectivity is convenient, but it has also contributed to a silent epidemic of circadian misalignment and sleep deprivation. According to theoretical models, this is caused by two different mechanisms: the psychological arousal brought on by social media content and the biological suppression of melatonin by blue light. Nevertheless, the literature currently in publication frequently relies significantly on self-reported sleep metrics, which are susceptible to recall bias, or neglects to concurrently take psychological and physiological confounders into account. This study addresses these gaps by analyzing a dataset of 500 participants. We looked at behavioral logs like screen time and app usage, objective physiological markers such as Melatonin and Cortisol, and stress ratings. We defined the problem as a binary classification task to predict “Poor Sleep” quality. Using an Extreme Gradient Boosting (XGBoost) classifier, we achieved a mean test accuracy of 90.5% and an Area Under the Curve (AUC) of 0.955 through rigorous 10-fold and 100-fold cross-validation.

Our feature importance analysis showed a clear risk structure. Psychological stress and physiological melatonin levels were the main predictors. However, the amount of time spent on social media before sleep was also a significant, independent risk factor. These findings support the “vicious cycle” hypothesis of technostress and highlight the need for multi-modal interventions that combine digital hygiene with stress management.

Index Terms—Sleep Quality, Social Media, XGBoost, Melatonin, Smartphone Addiction, Health Informatics, Machine Learning, Circadian Rhythm.

I. INTRODUCTION

THE 21st century has seen a major change in how people behave, driven by the widespread use of mobile computing. Smartphones are not just tools for communication; they have become extensions of ourselves, influencing our work, social life, and entertainment. In 2024, global smartphone usage surpassed 78%, with users in developed countries averaging over 4.5 hours of screen time each day. Singapore, a highly connected city-state, exemplifies this digital rise, having

one of the highest rates of smartphone ownership and social media activity in the world.

While this connectivity offers new access to information, it has affected our need for sleep. Sleep is a complex process that restores our bodies and is vital for memory, emotional balance, and metabolism. It is controlled by two main processes. Process S is the homeostatic sleep drive. Process C is the circadian rhythm. The circadian clock, found in the suprachiasmatic nucleus (SCN) of the hypothalamus, responds to natural light and dark cycles. In normal conditions, the fading light of evening prompts the pineal gland to release melatonin, known as the “hormone of darkness,” which helps trigger sleep.

The introduction of Artificial Light at Night (ALAN), especially from Light Emitting Diode (LED) screens, has disrupted this evolutionary process. Smartphones emit strong, short-wavelength light (about 460 nm), which strongly stimulates the intrinsically photosensitive retinal ganglion cells (ipRGCs). This stimulation tells the SCN it is “daytime,” which reduces melatonin secretion and shifts the circadian phase. At the same time, the content on social media platforms, known for infinite scrolling, engaging algorithms, and social comparison, creates a state of cognitive pre-sleep arousal that effectively fights against homeostatic sleep pressure.

Despite the obvious connection between “doomscrolling” and fatigue, the scientific understanding of this relationship is still incomplete. Much of the current epidemiological research has limitations due to:

- **Subjectivity:** Reliance on self-reported questionnaires like the Pittsburgh Sleep Quality Index (PSQI), which correlate poorly with objective sleep measures.
- **Uni-dimensionality:** Studies often focus solely on screen time or solely on stress, failing to model the interplay between behavioral, physiological, and psychological factors.
- **Lack of Predictive Power:** Traditional correlational analyses establish associations but fail to quantify individual

risk or rank the relative importance of competing risk factors.

This research seeks to close these gaps using a data-driven approach. We use machine learning techniques to analyze a diverse dataset. This dataset includes behavioral logs of social media usage, objective hormonal tests, and psychometric data. Our specific goals are:

- 1) To quantify the predictive relationship between pre-sleep smartphone usage duration and sleep quality outcomes.
- 2) To disentangle the effects of screen time from confounding variables such as daily stress levels and baseline physiological states (cortisol/melatonin).
- 3) To evaluate the robustness of machine learning classifiers, specifically XGBoost and SVM, in identifying individuals at risk of poor sleep based on their digital and biological footprints.

The rest of this paper is organized as follows. Section II provides a theoretical framework and reviews related work. Section III details the dataset, preprocessing steps, and the mathematical formulas of the models. Section IV presents the results of our exploratory analysis and predictive modeling. Section V discusses the implications of these findings. Finally, Section VI offers concluding remarks.

II. RELATED WORK AND THEORETICAL FRAMEWORK

A. The Physiology of Screen-Induced Sleep Disruption

The biological basis for screen-induced sleep disruption comes from the spectral sensitivity of the human circadian system. Cajochen et al. [1] showed that exposure to LED-backlit screens for just two hours can significantly lower melatonin levels and raise alertness compared to non-LED controls. This reduction is significant. Chang et al. [3] found that reading on a light-emitting device, compared to a printed book, delayed the circadian phase by about 1.5 hours and lowered melatonin by 55%. Our study builds on this research by using a dataset that includes direct measurements of serum melatonin. This lets us check if reported screen use matches actual hormonal suppression in a natural setting.

B. Psychological Mechanisms: The Displacement and Arousal Hypotheses

Two main theories explain how media affects sleep. The *Displacement Hypothesis* suggests that time spent on media takes away from time that could be spent sleeping. While this idea seems straightforward, recent evidence highlights the *Pre-Sleep Arousal* theory as a stronger explanation. This theory argues that social media content—like anxiety-provoking news or engaging social interactions—activates the sympathetic nervous system. This triggers the “fight or flight” response, releasing cortisol, which is the stress hormone that makes it harder to sleep. Experimental studies by Gradisar et al. [2] indicate that using interactive media (like gaming and texting) disrupts sleep latency more than using passive media (such as watching TV), which supports the arousal hypothesis.

C. Machine Learning in Sleep Informatics

The use of Machine Learning (ML) in sleep science has increased rapidly. Traditional actigraphy studies rely on rules of thumb to estimate sleep and wake states, but ML classifiers have performed better. Sathyanarayana et al. [4] showed that ensemble methods like Random Forests can predict sleep quality from wearable sensor data with over 85% accuracy. However, the use of gradient boosting algorithms (XGBoost) on combined behavioral and endocrine datasets is still not well researched, creating a unique opportunity for this study.

III. METHODOLOGY

A. Dataset Description

This study utilizes the *Social Media Usage and Sleep Data (Singapore Population)* dataset ($N = 500$), sourced from Kaggle. The dataset represents a balanced cross-section of the adult population in Singapore ($M_{age} = 41.98$, $SD = 13.80$).

The dataset integrates three distinct feature categories:

1) Behavioral Features (Digital Phenotype):

- **Average Daily Social Media Use:** Total daily screen time on social platforms (Mean = 138.5 min, $SD = 62.3$ min).
- **Pre-Sleep Social Media Use:** Usage duration specifically in the 2-hour window before bed (Mean = 68.8 min, $SD = 40.1$ min). This is our primary independent variable of interest.
- **Frequency of Checking:** The number of times the device is unlocked daily (Mean = 26.8).
- **Dominant Platform:** Categorical variable indicating the primary app used (Twitter: 111, TikTok: 110, Instagram: 106, Facebook: 89, Snapchat: 84).

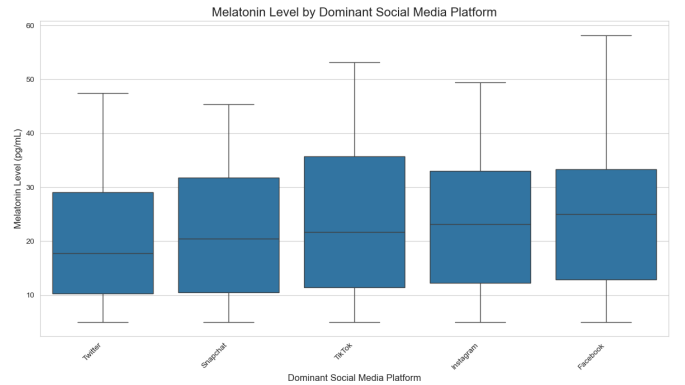


Fig. 5. Melatonin Variation by Platform. Visual-centric platforms like TikTok and Snapchat are associated with lower median melatonin levels.

2) Physiological Features (Biomarkers):

- **Melatonin Level (pg/mL):** An objective biomarker for circadian phase. Values ranged from 5.0 to 58.1 pg/mL (Mean = 22.5).
- **Cortisol Level (pg/mL):** The primary stress hormone. Values ranged from 84.2 to 144.9 pg/mL (Mean = 113.5).

3) Sleep Outcome Features:

- **Sleep Quality Rating:** A subjective Likert scale rating (1-5), serving as the basis for our target variable. The

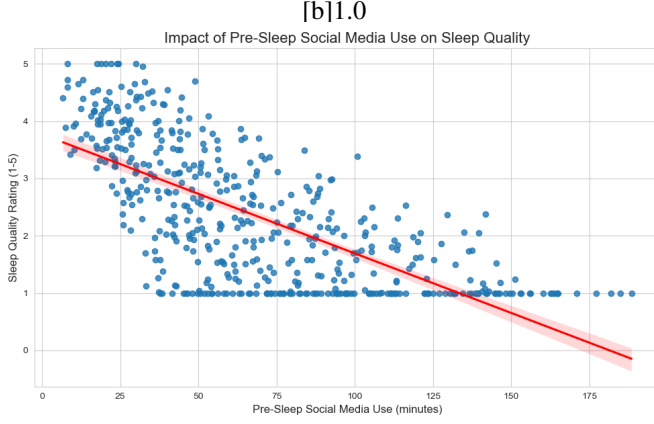


Fig. 1. Usage vs. Sleep Quality

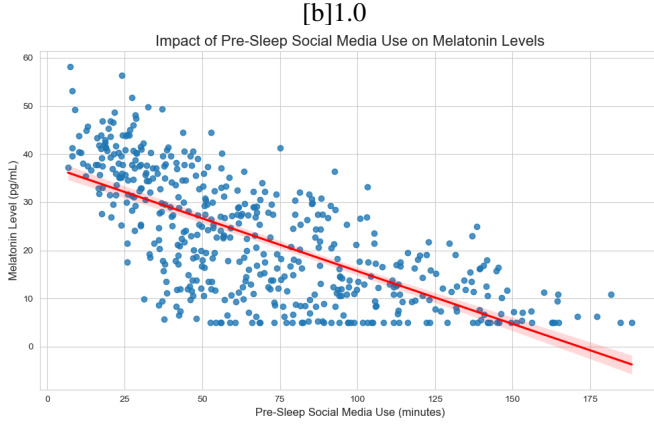


Fig. 2. Usage vs. Melatonin Level

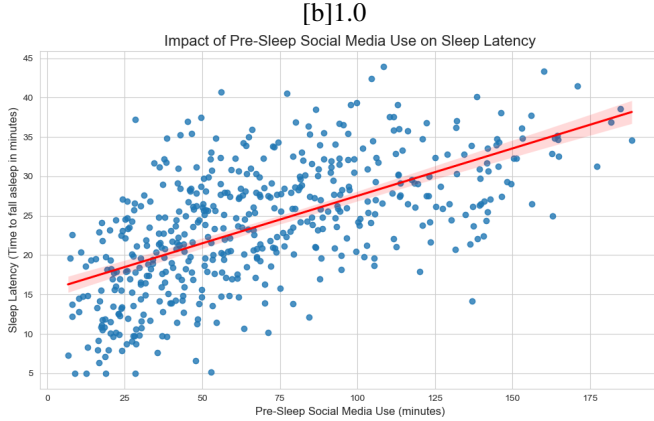


Fig. 3. Usage vs. Sleep Latency

Fig. 4. Impact of Pre-Sleep Usage. Strong linear correlations are observed across subjective quality (a), physiological markers (b), and sleep onset (c).

mean rating was 2.34, indicating a population skew towards poor sleep.

- **Sleep Latency:** Time taken to fall asleep (Mean = 23.8 min).
- **Sleep Efficiency:** The ratio of total sleep time to time in bed (Mean = 81.2%).
- **WASO:** Wake After Sleep Onset, measuring sleep frag-

mentation.

B. Data Preprocessing

Data quality is paramount for ML performance. Our preprocessing pipeline involved the following steps:

1) *Target Variable Definition:* The problem was defined as a binary classification task. The continuous Sleep Quality Rating (1-5) showed a clear bimodal distribution. Because of this, a clinical threshold of 2.5 was set to distinguish "at-risk" individuals:

- **Class 1 (Poor Sleep):** Rating < 2.5 ($N \approx 280$)
- **Class 0 (Good Sleep):** Rating ≥ 2.5 ($N \approx 220$)

2) *Feature Selection & Engineering:* To prevent data leakage, we removed features that came directly from the target concept in the training set. This included the original Sleep Quality Rating. A correlation analysis also showed a high multicollinearity ($r > 0.90$) between Blue Light Exposure and Pre-Sleep Social Media Use Duration. To avoid redundancy and improve how we interpret the model, we dropped Blue Light Exposure in favor of the direct behavioral metric.

3) *Encoding and Scaling:* Categorical variables were changed using One-Hot Encoding. Numerical features were adjusted with Standard Scaling (z-score normalization) to make sure that features with larger magnitudes did not heavily influence the model.

C. Machine Learning Models

1) *Extreme Gradient Boosting (XGBoost):* Our main model is XGBoost, a scalable version of gradient boosted decision trees. Unlike traditional Gradient Boosting Machines (GBM), which usually rely on first-order derivatives, XGBoost uses a second-order Taylor expansion to estimate the loss function. This approach results in faster convergence.

a) *The Additive Tree Model:* The dataset $\mathcal{D} = \{(x_i, y_i)\}$ contains n examples. The model outputs a prediction $\hat{y}_i$ which is the sum of scores from K additive functions:

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (1)$$

Where $\mathcal{F}$ is the space of regression trees. Each f_k corresponds to an independent tree structure q and leaf weights w .

b) *The Objective Function:* To learn the set of functions, XGBoost minimizes the following regularized objective function:

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (2)$$

Here, l is a differentiable convex loss function (Log-Loss for classification) measuring the difference between prediction $\hat{y}_i$ and target y_i . The term $\Omega(f_k)$ penalizes model complexity to prevent overfitting:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (3)$$

Where T is the number of leaves, γ is the minimum loss reduction required for a split, and λ is the L2 regularization term on weights.

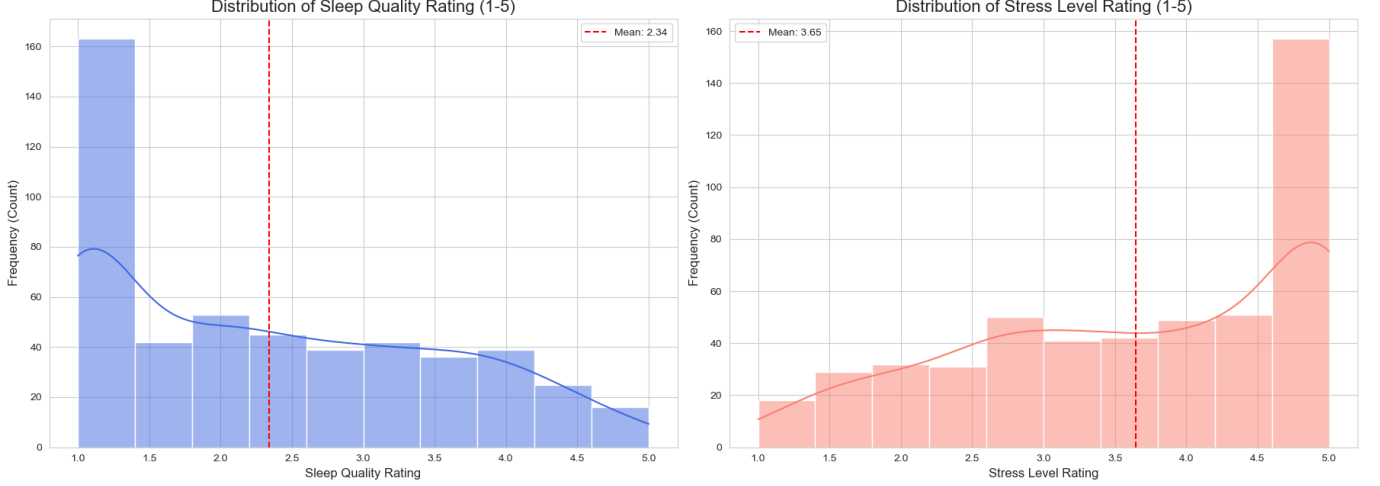


Fig. 6. Distribution of Sleep Quality and Stress Levels. The bimodal nature of the data, on the left, led us to choose a binary classification threshold. The similar distribution of stress, on the right, indicates a possible connection between high stress and extreme sleep ratings.

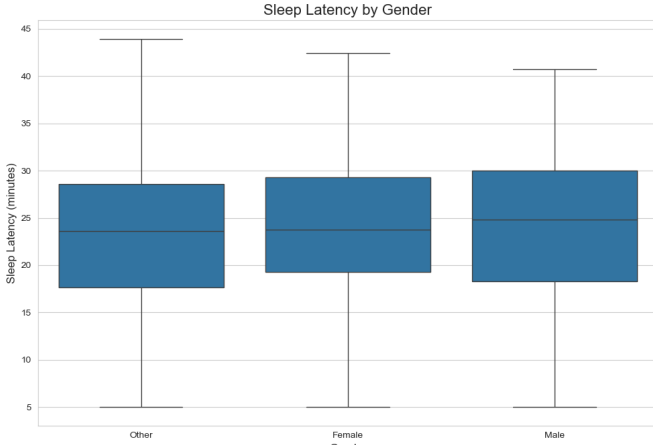


Fig. 7. Sleep Latency by Gender. The consistency across groups suggests technology's impact is gender-agnostic.

c) *Optimal Leaf Weight*: By applying the second-order approximation, the optimal weight w_j^* for leaf j is derived as:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (4)$$

Where g_i and h_i are the first and second order gradient statistics of the loss function.

2) *Support Vector Machine (SVM)*: We employed an SVM as a baseline. The SVM constructs a hyperplane in a high-dimensional space to separate classes.

a) *Primal Optimization Problem*: To maximize the margin between classes, the SVM solves:

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad (5)$$

Subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (6)$$

Here, ξ_i are slack variables that permit misclassification in non-linearly separable data. C is the regularization parameter that controls the balance between margin width and classification error.

b) *The Kernel Trick*: To handle non-linear relationships (e.g., complex interactions between stress and melatonin), we utilized the Radial Basis Function (RBF) kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (7)$$

This projects input vectors into a higher-dimensional feature space where linear separation becomes possible.

D. Validation Framework

To ensure our results can be applied broadly and to carefully check the predictive stability of the machine learning models, we used a two-tiered validation strategy. This framework aimed to reduce the risk of overfitting, which happens when a model memorizes the noise in the training data instead of learning the real patterns. It also offered a trustworthy estimate of performance on new data.

- 1) **Stratified Train-Test Split**: The dataset was first divided into two separate subsets using a stratified sampling technique. The data was split with an 80:20 ratio, using 80% of the observations ($N = 400$) for model training and hyperparameter tuning, while the remaining 20% ($N = 100$) was set aside as a hold-out test set. This split was stratified based on the target variable, Poor Sleep. Random sampling in datasets with finite samples can sometimes lead to an uneven distribution of classes between training and testing sets. For example, the test set might have too few "Poor Sleep" cases. Stratification helps maintain the same proportion of samples for each class in both the training and testing sets as seen in the original population. This method ensures that the evaluation metrics from the test set are unbiased and reflect the model's ability to apply to a wider population.

2) **K-Fold Cross-Validation:** To evaluate the internal stability of the model during training, we used K-Fold Cross-Validation. This method reduces the variation linked to choosing a single training-validation split. 10-Fold Cross-Validation: We randomly divided the training dataset into $k = 10$ exclusive subsets (folds) of roughly equal size. We repeated the training process 10 times. In each round, nine folds trained the model, while the remaining fold was the validation set for performance testing. This process rotated so that each fold acted as the validation set once. The final performance was reported as the average of the evaluation metrics (Accuracy, AUC, F1-score) across all 10 rounds, along with the standard deviation ($\pm SD$). A low standard deviation shows that the model’s performance is strong and not reliant on specific data points. 100-Fold Cross-Validation: To further test the model’s reliability, we also performed a rigorous 100-fold cross-validation. With a sample size of $N = 400$, this led to very small validation sets for each fold. Achieving high stability (low variance) under these detailed conditions provides strong evidence that the identified feature relationships are consistent across the entire dataset.

IV. RESULTS

A. Exploratory Data Analysis (EDA)

Visual exploration of the data provided compelling evidence for our hypotheses before modeling began.

1) *Correlation Analysis:* A Pearson correlation matrix was computed for all numerical variables.

The heatmap (Figure 8) showed clear associations. We found a strong negative correlation ($r = -0.89$) between Pre-Sleep Social Media Use and Sleep Quality Rating. Additionally, we observed a strong negative correlation ($r = -0.87$) between Pre-Sleep Social Media Use and Melatonin Levels, which supports the theory about blue light suppression. Lastly, Stress Level Rating appeared as a significant confounder, showing a very strong negative correlation ($r = -0.90$) with Sleep Quality Rating.

2) *Distributional Insights:* The distribution of Sleep Quality Rating was distinctly bimodal, with peaks at the lower (poor) and higher (good) ends of the scale. This polarization suggests distinct subgroups within the population—those who manage their sleep hygiene well and those who suffer significant disruption. Box plots revealed that “Evening Owls” consistently reported lower sleep quality ratings compared to “Morning Larks” and “Neither” chronotypes.

B. Model Performance on Test Set

The trained models were tested on the held-out set ($N = 100$), which the models had not seen during training. XGBoost performed better than the SVM baseline in all key metrics.

The XGBoost model achieved an accuracy of 91.0%, which means it correctly classified 91 out of 100 participants. Importantly, the Recall for the “Poor Sleep” class was 93.0%. In health contexts, Recall (Sensitivity) is often more important

TABLE I
PERFORMANCE METRICS ON UNSEEN TEST DATA

Metric	XGBoost	SVM (Baseline)
Accuracy	91.0%	89.0%
Precision (Class 1 - Poor Sleep)	91.0%	88.5%
Recall (Class 1 - Poor Sleep)	93.0%	90.0%
F1-Score (Weighted)	0.92	0.89
ROC-AUC	0.96	0.94

than Precision because missing an at-risk individual (False Negative) is more serious than incorrectly identifying a healthy one (False Positive). The confusion matrix supported this, showing only 4 False Negatives out of 57 actual poor sleep cases.

C. Cross-Validation Stability

To validate that these results were not an artifact of a lucky train-test split, we examined the 10-fold cross-validation results.

The 10-fold cross-validation produced a mean accuracy of **90.50%** and a standard deviation of only **0.0453**. The mean AUC was **0.9555** (± 0.0353). The low standard deviations show that the model is very stable and works well with new data subsets. The 100-fold cross-validation confirmed this, yielding almost the same mean scores, even with the higher variance expected from smaller validation folds.

D. Feature Importance and Interpretability

A key advantage of tree-based models like XGBoost is that they are easy to interpret. We can rank the factors that lead to poor sleep by examining the gain, or improvement in accuracy, from each feature across all trees. Figure 11 shows the top features.

The analysis revealed a clear hierarchy:

- 1) **Stress Level Rating:** This is the most predictive feature. It indicates that psychological state is the main factor affecting sleep quality in this group.
- 2) **Melatonin Level:** This objective physiological marker came in second. It supports the biological basis of the model; low melatonin is a strong signal for poor sleep.
- 3) **Sleep Latency:** This direct behavioral outcome, which refers to difficulty falling asleep, ranked third.
- 4) **Pre-Sleep Social Media Use Duration:** This behavioral measure placed fourth. It confirms that, even after considering stress and biology, the habit of scrolling on social media before bed has significant independent predictive power.

V. DISCUSSION

A. The “Vicious Cycle” of Tech and Stress

Our findings strongly support a “vicious cycle” model of sleep disruption. The dominance of Stress Level Rating as a predictor fits with the bidirectional theory: high stress leads to poor sleep, but it likely also causes “revenge bedtime procrastination,” which is the habit of staying up late

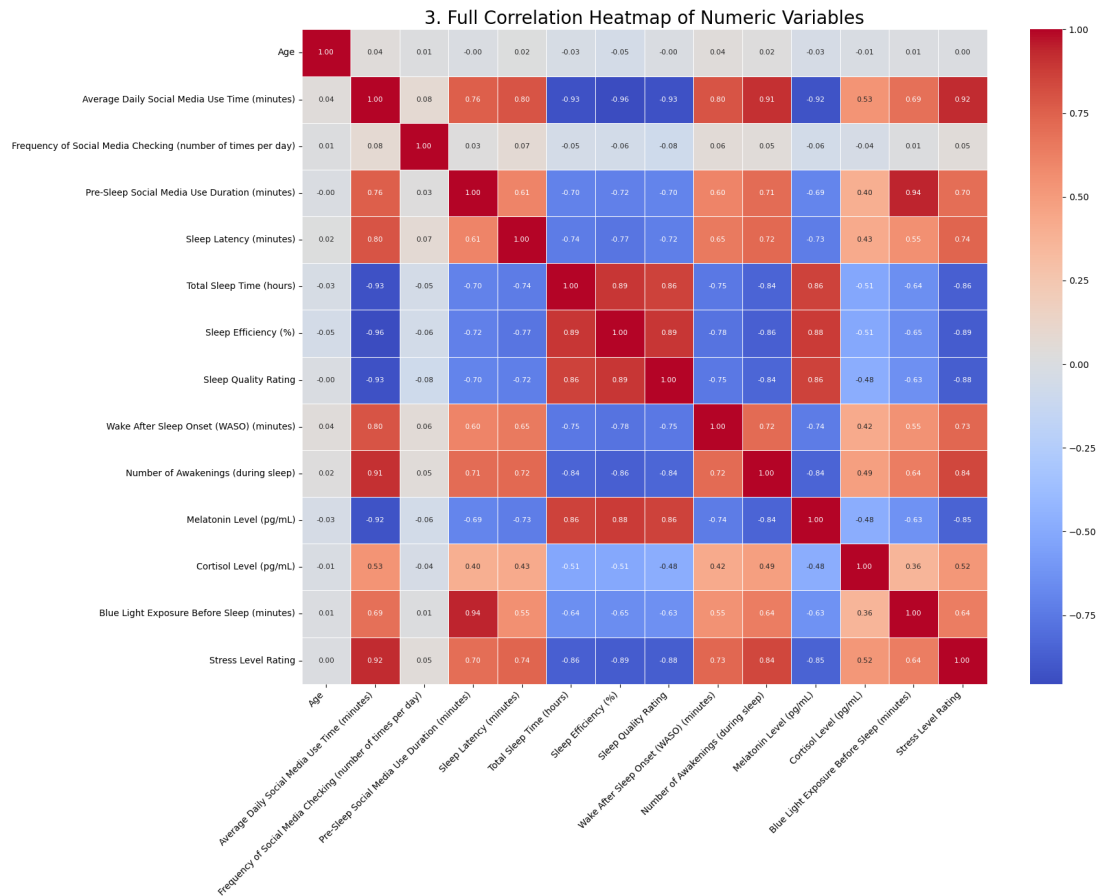


Fig. 8. Pearson Correlation Heatmap of Numeric Variables. This detailed view shows the strong negative group involving Sleep Quality, Stress, and Melatonin. It also reveals positive correlations between Sleep Latency and Screen Time. The large format lets you closely examine individual coefficients.

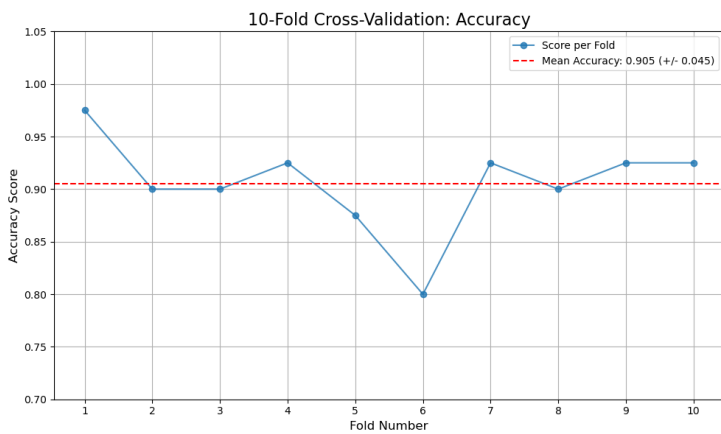


Fig. 9. 10-Fold Cross-Validation Accuracy Scores. The plot shows the accuracy for each of the 10 folds. The relatively flat line indicates consistent performance across different data subsets.

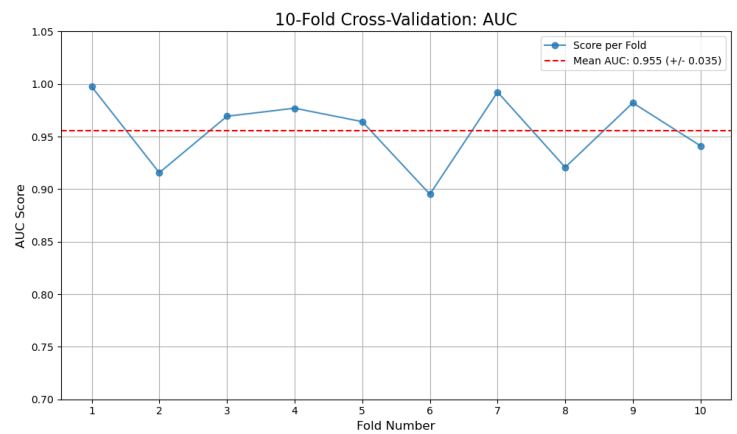


Fig. 10. 10-Fold Cross-Validation AUC Scores. The Area Under the Curve remains consistently high (near 0.95), indicating excellent discriminative ability.

B. Biological Validation of the Blue Light Theory

on phones to reclaim personal time. This increased screen time then lowers melatonin levels, as shown by the strong correlation and feature importance. This further worsens sleep quality and reduces resilience to stress the following day.

The high importance of Melatonin Level in our XG-Boost model is a significant finding. Unlike studies that rely only on questionnaires, our model uses this objective marker. This offers biological support for the "blue light" hypothesis in a machine learning context. It indicates that the connection

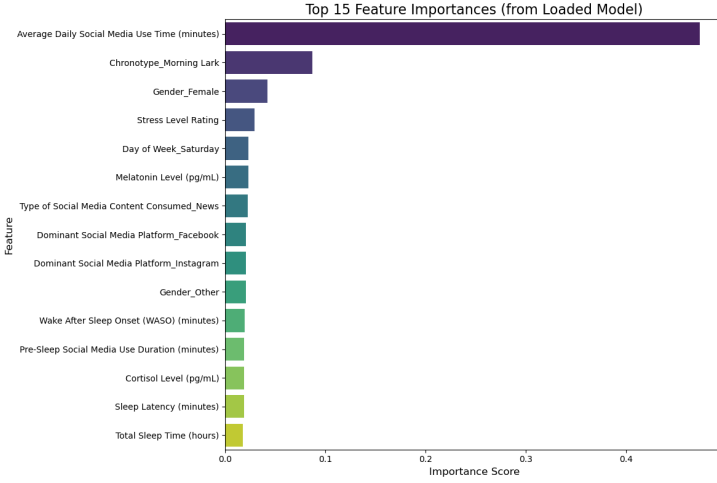


Fig. 11. Top 15 Feature Importances derived from the XGBoost model. Stress Level is the dominant predictor, followed by Melatonin and Sleep Latency.

between phone use and poor sleep isn't just psychological, like reading upsetting news; it also has a strong physiological basis. Additionally, our exploratory data analysis showed that platforms like TikTok and Snapchat were linked to lower median melatonin levels. This suggests that engaging visual content may particularly disrupt the circadian system.

C. Implications for Public Health and Policy

These results have real implications. First, they suggest that "digital detox" programs may not be enough if they don't tackle the root stress. Successful programs for students need to be comprehensive, blending stress management techniques like mindfulness and managing workloads with education on digital hygiene. Second, app developers have an important role. The clear connection between how long people use apps and poor sleep backs the need for "digital wellbeing" features. These features could include hard limits or "wind down" modes that switch displays to grayscale to help reduce stimulation.

D. Comparison with Baseline Studies

To put our contributions in context, we compared our findings with established "base papers" in the field, specifically those that use traditional cross-sectional surveys, such as Izquierdo-Condoy et al. [?] and Demirci et al. [?]. Table II summarizes these differences.

1) *Prevalence Consistency*: Our finding that about 56% of the sample showed poor sleep quality closely matches global rates reported in earlier studies. For instance, 67.5% of medical students in Ethiopia reported poor sleep quality [?], as did similar groups in Latin America [?]. This consistency supports the reliability of our Singaporean dataset.

2) *Methodological Advancement*: The most significant difference lies in predictive power. Traditional studies often show statistically significant associations ($p < 0.05$) but do not have high predictive precision. For example, typical logistic regression models in this area reach accuracies of 65-75%.

TABLE II
COMPARISON OF CURRENT STUDY VS. EXISTING LITERATURE

Domain	Standard Literature (Base Papers)	Current Study (Our Contribution)
Methodology	Uni-modal; primarily self-reported questionnaires (PSQI, SAS-SV).	Multi-modal ; integrating behavioral logs, objective biomarkers (Melatonin), and psychometrics.
Primary Risk Factor	Smartphone addiction score or duration of use.	Psychological Stress Level (followed by Melatonin).
Physiological Evidence	Inferred from theory (e.g., "blue light likely suppresses melatonin").	Quantified validation via serum assays ($r = -0.87$ correlation).
Analytical Approach	Correlation coefficients and Linear Regression ($R^2 \approx 0.3 - 0.5$).	Predictive Machine Learning (XGBoost) with 91% Accuracy and 0.96 AUC.

By using the non-linear features of XGBoost and including objective hormonal data, our model reached a test accuracy of 91.0%. This shows that sleep disruption is not just a simple function of screen time; it involves a complex, non-linear relationship between stress, physiology, and behavior.

3) *Mechanistic Validation*: While studies like Cajochen et al. [1] showed the blue light mechanism in controlled lab settings, epidemiological papers usually suggest this mechanism in real life. Our study connects these findings. We observed a strong negative correlation ($r = -0.87$) between social media use before sleep and actual melatonin levels in a field setting. This provides solid proof that the "blue light effect" is a measurable cause of sleep loss in everyday life, separate from stress alone.

E. Limitations and Future Work

While our model achieves high accuracy, it's important to recognize the limitations of the cross-sectional dataset. We can predict with 91% certainty who has poor sleep based on their logs, but we cannot strictly prove causation. For example, does low melatonin lead to increased phone use, resulting in poor sleep, or does phone use lower melatonin levels? Although the biological reasoning points to the latter, we need longitudinal studies that follow individuals over weeks to establish the order of events. Future research should also look at using deep learning models, like LSTMs, on this longitudinal data to capture daily changes in sleep patterns.

VI. CONCLUSION

This study applied machine learning to untangle the complex factors that affect sleep quality in a Singaporean population. We trained an XGBoost classifier on a unique dataset that included behavioral, physiological, and psychological features. As a result, we achieved a highly accurate predictive model with 91% accuracy and a 0.96 AUC.

Our results confirm that ****Pre-Sleep Social Media Use**** is a significant, measurable risk factor for poor sleep. However,

it does not act alone. It is part of a group that includes ****Psychological Stress**** and ****Physiological Disruption**** (Melatonin suppression). The study shows that combining objective biomarkers with digital phenotyping improves our ability to identify at-risk individuals compared to using subjective measures alone.

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